

Transfer Attacks in Face Recognition

OPS: Operator-Perturbation-based Stochastic Optimization

Boosting Adversarial Transferability through Augmentation in Hypothesis Space - CVPR 2025

Name-Kkartik Aggarwal

Github: https://github.com/skigeek16/Transfer_attack

Attacks Considered

Evaluation includes a proposed OPS method and five transfer-attack baselines.

Attack	Nature in this evaluation	Role
PGD	Projected gradient baseline	Baseline
MI-FGSM	Momentum iterative FGSM	Baseline
TI-FGSM	Translation-invariant gradients	Baseline
SI-NI-FGSM	Scale-invariant + Nesterov momentum	Baseline
MI-ADMIX-DI-TI	Momentum + admix + DI + TI	Strong baseline
OPS	Operator-perturbation stochastic optimization	Proposed

Why these baselines matter

PGD establishes a direct iterative L-infinity reference point.

Momentum, input diversity, translation invariance, scale invariance, and admix test common transferability improvements.

OPS is compared against all baselines on the same subset and victim-model pool.

OPS Definition, Nature, Basic Idea, and Face-Verification Goal

Definition & nature

Transfer-based, gradient-based adversarial attack.

L-infinity bounded perturbation method.

Optimizes in embedding space for face verification.

Basic idea

Sample nearby perturbed images around the current adversarial point.

Apply stochastic operator sequences to each sample.

Average gradients over the transformed batch to improve transferability.

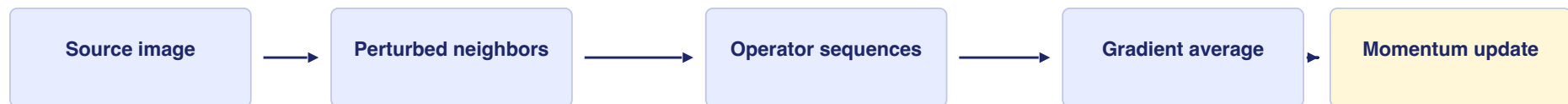
Goal in face verification

Objective uses cosine similarity in embedding space.

Impersonation: increase similarity with the target identity.

Dodging: decrease similarity with the target identity.

Hypothesis-space augmentation view



Operators: identity, flips, shifts, rotations, intensity scaling, DIM-style resize-padding

Working of OPS Attack

Numbered procedure

- 1 Initialize**
Start from the original image; initialize normalized gradient momentum.
- 2 Sample neighbors**
Draw nearby perturbations using OPS_BETA and OPS_SAMPLE_LEVELS.
- 3 Sample operators**
Select stochastic operator sequences for each neighbor.
- 4 Build batch**
Apply operators to $adv + noise$ and concatenate transformed samples.
- 5 Compute objective**
Use cosine-similarity loss against the target embedding.
- 6 Average gradients**
Backpropagate through the transformed batch and aggregate gradients.
- 7 Momentum update**
Normalize gradients, update accumulated momentum, and step by $\text{sign}(g)$.
- 8 Project & clip**
Project to the epsilon ball and clip to the valid pixel range.

Operator family used in OPS

identity

flips

shifts

rotations

intensity scaling

DIM resize-padding

OPS Implementation Details in This Repository

Integration checklist

Item	Status
OPS added in ALL_ATTACKS	Done
OPS: ops_path added in ATTACK_COLS	Done
OPS function implemented inside core/transfer_attack_core.py	Done
OPS connected through run_attack(...)	Done
Final attack run used only docs/subset_input_pairs.csv	Done

Hyperparameters

Parameter	Value
EPSILON	0.062
NUM_ITER	5
DECAY	1.0
OPS_BETA	2.0
OPS_NUM_NEIGHBOR	10
OPS_NUM_OPERATOR	10
OPS_SAMPLE_LEVELS	(2, 3, 4)

Core control path

```
ALL_ATTACKS += ['OPS'] | ATTACK_COLS['OPS'] = 'ops_path'
run_attack('OPS', ...) -> ops_attack(...) -> stochastic neighbor/operator batch -> momentum update
```

Implementation file: `core/transfer_attack_core.py`

Experimental Setup

Evaluation subset and metrics

Subset: docs/subset_input_pairs.csv.

Total evaluated rows: 480.

Self-transfer pairs excluded.

Metrics: breach success rate and mean embedding impact.

Model pool

Role	Models
Surrogate attacker	Facenet512, ArcFace, GhostFaceNet, VGG-Face
Victim	Facenet512, ArcFace, GhostFaceNet, VGG-Face, IR152

Evaluation matrix



Each non-self surrogate-victim pair is evaluated under dodging and impersonation goals.

Results and Conclusion

Overall performance

Attack Method	Breach Rate (%)	Mean Impact
OPS (Proposed)	40.42	0.2258
SI-NI-FGSM	29.17	0.1755
MI-FGSM	26.67	0.1643
MI-ADMIX-DI-TI	24.17	0.1535
TI-FGSM	20.42	0.1313
PGD	16.67	0.1003

By attack goal

Goal	Breach Rate	Mean Impact
Dodging	52.08%	0.2946
Impersonation	28.75%	0.1571

Breach-rate comparison

OPS		40.42%	0.2258
SI-NI-FGSM		29.17%	0.1755
MI-FGSM		26.67%	0.1643
MI-ADMIX-DI-TI		24.17%	0.1535
TI-FGSM		20.42%	0.1313
PGD		16.67%	0.1003

Best surrogate-victim pairs

Pair	Breach	Impact
Facenet512 -> ArcFace	63.33%	0.2547
Facenet512 -> VGG-Face	60.00%	0.2399
VGG-Face -> ArcFace	60.00%	0.2500
VGG-Face -> Facenet512	60.00%	0.4786
ArcFace -> VGG-Face	53.33%	0.2016
ArcFace -> GhostFaceNet	50.00%	0.1525

Conclusion: OPS produces the highest breach rate and mean embedding impact, indicating stronger transferability than all tested baselines on the evaluated subset.